
Project Setu: Master Plan V4.0 (Finalized)

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Status: Architecture Finalized (Pre-Development) — May 2026

Primary Goal: Provide a high-performance, mobile-first, and "almost free" utility for EAPCET students.

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1. Executive Summary & Context

Project Setu is an intelligent ecosystem designed to navigate the "Counseling Chaos" for over 500,000 students in the MPC and BiPC streams. It replaces fragmented PDF data with a structured, agentic advisor.

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2. The 3-Tier Intelligence Architecture

To ensure the project is fast and costs nearly nothing to run, we have segregated tasks into three layers:

Tier 1: The Engine (Deterministic SQL)

- **Tech:** Supabase (Postgres) + Next.js.
- **Goal:** 100% numerical accuracy and $0 API cost.
- **Features:**
  - **The Probabilistic Predictor:** Instantly categorizes college branches into **Safe, Probable, and Dream**.
  - **Phase-Gap Seat Matrix:** Real-time updates on seat vacancies between rounds.
  - **Fee Transparency:** Displays official AFRC-sanctioned fees for Management Quota.
  - **College Directory:** Comprehensive list of branches and intake capacities.

Tier 2: The Librarian (Standard RAG)

- **Tech:** Gemini 1.5 Flash + Context Caching.

- **Goal:** Interpreting text-heavy rules with low latency.
- **Features:**
  - **Rulebook AI:** Answers questions about eligibility and fee reimbursement using official manuals.
  - **Branch Matchmaker:** Provides personalized summaries after the initial profiling quiz.

Tier 3: The Architect (Agentic LangGraph)

- **Tech:** LangChain + LangGraph.

- **Goal:** Stateful journey management and complex reasoning.

- **Features:**
  - **Verification Center Navigator:** An interactive "Day in the Life" guide to physical verification.
  - **Web Option Strategy Guard:** An autonomous agent that reviews the entire web options list for strategic errors.

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3. The Predictor Logic (Tier 1 Math)

Predictions are calculated using a 3-year weighted average (C_{avg}) of closing ranks:

- **Safe:** $R < (C_{avg} \times 0.85)$
- **Probable:** $(C_{avg} \times 0.85) \le R \le (C_{avg} \times 1.10)$
- **Dream:** $R > (C_{avg} \times 1.10)$

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4. Technical Stack Alignment

- **Frontend:** Next.js (Mobile-first PWA for rural accessibility).
- **Database:** Supabase (PostgreSQL).
- **AI Models:** Gemini 1.5 Flash (for high-speed RAG) and Gemini 1.5 Pro (for complex LangGraph reasoning).

- **Hosting:** Vercel (with Edge Caching to survive result-day traffic spikes).

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5. Operational Security

- **No Hallucination Policy:** Numerical data is **never** generated by AI; it is pulled directly from SQL.
- **Rate Limiting:** IP-based "Token Buckets" to protect free API quotas.

- **Data Integrity:** Automated validation scripts flag any anomalies in scraped PDFs before they are added to the database.
